Research mapping of IISER Bhopal, Mohali and Thiruvananthapuram, 2016–2026

DPCN course project • Group 20, team **EigenMinds**: Ponnambalam V, Manit Roy, Shashwat Mukadam • First-round report • Instructor: Prof. Chittaranjan Hens

Interactive dashboard: `(linkaddedondeployment)` Code, data and network files: `(linkaddedondeployment)`

Key findings.

- **Same quality, different growth.** The three IISERs publish an almost identical share of Q1 papers (73.5%, 74.2%, 73.7%; χ^2 $p = 0.85$, Cramér's $V = 0.014$), but output grows at very different rates: 16.7% a year at TVM, 12.2% at Bhopal and 8.1% at Mohali (2016→2025).
- **A few giant papers carry the citations.** 161 papers with more than 50 authors (Planck, pulsar timing, collider experiments) hold a fifth of all citations; TVM loses almost half its citation count without them (53,136 → 27,441). Field-weighted impact (FWCI) is the fair comparison, and there Bhopal leads (median 0.88 vs 0.81 and 0.73; Kruskal–Wallis $p = 0.00055$).
- **Three islands.** 96% of citations between the three institutes' papers stay inside one institute, and only 29 of 304 faculty co-authorship ties cross between them. Collaboration communities follow the institute (assortativity 0.85) more than the department (0.48).
- **Net importers of Indian knowledge.** Each IISER cites other Indian institutes more than they cite it back (in/out ratio 0.59, 0.70, 0.97), with TVM nearly in balance.

1 Data and method

Faculty. We scraped the current regular faculty (Assistant, Associate and full Professors) from each institute's own website: Bhopal's ten department sites, Mohali's *People* page and TVM's seven school pages. That gives 410 faculty (Bhopal 160, Mohali 118, TVM 119 matched). Visiting, adjunct, emeritus and fellowship posts are excluded so the institutes are counted alike. Each name was matched to an OpenAlex author profile by downloading every author OpenAlex has ever affiliated with the institute and fuzzy-matching locally, requiring both surname and given name to agree and preferring the profile whose field fits the department; 397 (96.8%) matched.

Papers. We fetched all 2016–2026 journal and conference papers of the matched authors from OpenAlex: 11,639 unique papers, giving 12,277 faculty–paper rows in `group20_data.csv`. Because OpenAlex lists truncate authors at 100, every larger paper was re-fetched in full. A row is *affiliated* when that faculty member's authorship lists the IISER (67% of rows); institute statistics count each affiliated paper once. Journal quartiles come from SCImago, matched on ISSN and year (93.3% of journal papers covered), and conference ranks from CORE 2023. For the Indian-connection columns we fetched the institutions of all 385k referenced works and every Indian-authored work citing a corpus paper.

Limits. Only current faculty are listed, so papers by people who left before 2026 are missing and institute totals undercount total institutional output. Citation counts are as of October 2026.

2 Output, quality and people

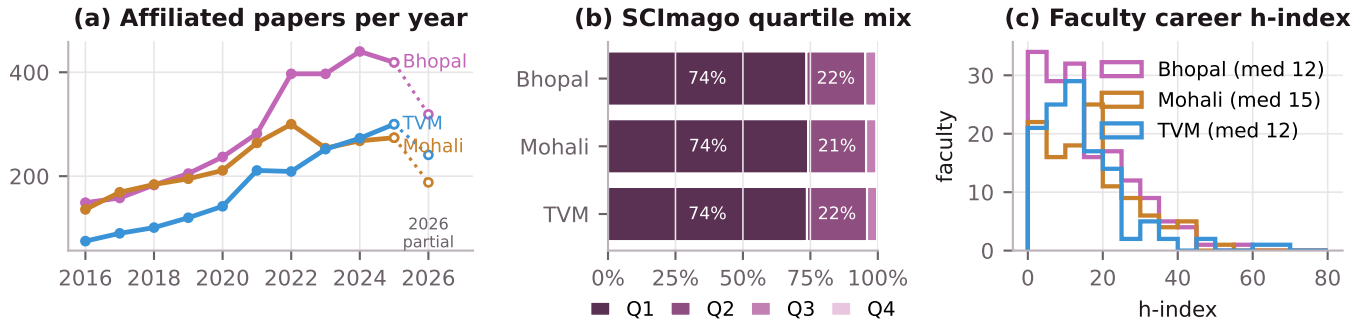


Figure 1: (a) Unique affiliated papers per year; 2026 is partial. (b) SCImago quartile mix of journal papers. (c) Career h-index of current faculty (OpenAlex), with medians.

Output rose at all three institutes (Mann–Kendall trend increasing, increasing, increasing). A negative-binomial regression of yearly counts on year with a 2020–21 indicator finds significant growth everywhere but *no* COVID effect at any institute (all $p > 0.5$). Bhopal's jump after 2021 is broad-based: Chemistry, Physics, EECS and Earth Sciences all grew by half or more between 2019–21 and 2022–24. The quartile mix is statistically indistinguishable across institutes (Table 1, Fig. 1b), and so is faculty h-index (Kruskal–Wallis $H = 3.44$, $p = 0.179$). The institutes differ in *concentration* instead: TVM's citation Gini of 0.82 reflects 36 large-collaboration papers, mostly Planck. Larger teams are cited more (Spearman $\rho = 0.20$ between author count and citations), which is one reason raw counts mislead.

Table 1: Institute-level statistics (unique affiliated papers, 2016–2026).

	Bhopal	Mohali	TVM	All three
Papers	3,186	2,443	2,014	7,597
Citations (excl. >50 authors)	52,576 (52,197)	43,294 (39,210)	53,136 (27,441)	148,485 (118,327)
Q1 share	73.5%	74.2%	73.7%	73.7%
Median FWCI / top-10% cited	0.88 / 16.5%	0.81 / 14.7%	0.73 / 14.8%	0.81 / 15.5%
NB-GLM growth per year (p)	14.7% (0.000)	8.0% (0.003)	17.7% (0.000)	13.1% (0.000)
COVID-years effect (p)	-3.5% (0.84)	9.4% (0.63)	11.6% (0.56)	3.7% (0.84)
International co-authorship / OA	26% / 44%	27% / 48%	30% / 41%	27% / 45%
Citation Gini	0.65	0.66	0.82	0.72
Median faculty h-index	12	15	12	–

3 Journals, research areas and most-cited papers

Table 2: Top 5 journals per institute (papers, SCImago quartile).

	Journal	n	Q
Bhopal	Physical Review B	122	Q1
	Chemical Communications	84	Q1
	Chemistry - A European Journal	63	Q1
	Organic Letters	61	Q1
	Chemistry - An Asian Journal	45	Q2
Mohali	Physical Review D	96	Q1
	Physical Review B	41	Q1
	Chemistry - A European Journal	41	Q1
	Chemical Communications	40	Q1
	Scientific Reports	39	Q1
TVM	Angewandte Chemie	58	–
	Angewandte Chemie International Edition	55	Q1
	Physical Review B	53	Q1
	Chemical Communications	42	Q1
	Chemistry - A European Journal	40	Q1

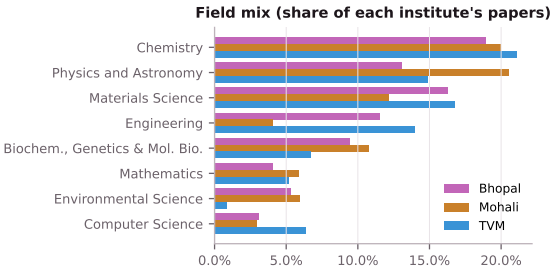


Figure 2: Research-area breakdown: share of each institute's papers in the eight largest OpenAlex fields.

Chemistry is the largest field at Bhopal and TVM (about a fifth of papers) and level with physics at Mohali, the most physics-heavy of the three. Bhopal and TVM put more into materials science and engineering. Earth and environmental science is about a tenth of Bhopal's and Mohali's output but under 2% of TVM's (Fig. 2). The physics venues (*Physical Review B* and *D*) and the chemistry venues (*Angewandte*, *Chem. Commun.*, *Chem. Eur. J.*) dominate the publication lists (Table 2).

Table 3: The ten most-cited papers (all three IISERs, citations to October 2026). Seven are parts of the *Planck 2018* release; excluding papers with more than 50 authors, the top paper is a 2020 *Advanced Functional Materials* paper on post-synthetic modification of metal–organic frameworks (dashboard: Journals & Impact, "$\leq$ 50 authors").

#	Title	Venue	Year	Cites	DOI
1	Planck 2018 results	Astronomy and Astrophysics	2020	14,257	10.1051/0004-6361/201833910
2	Planck 2018 results	Astronomy and Astrophysics	2019	2,426	10.1051/0004-6361/201833887
3	Planck 2018 results	Astronomy and Astrophysics	2021	1,855	10.1051/0004-6361/201833910e
4	Planck 2018 results	Astronomy and Astrophysics	2019	1,636	10.1051/0004-6361/201833880
5	Planck 2018 results	Astronomy and Astrophysics	2020	1,192	10.1051/0004-6361/201936386
6	Planck 2018 results	Astronomy and Astrophysics	2019	781	10.1051/0004-6361/201833886
7	Planck 2018 results	Astronomy and Astrophysics	2020	696	10.1051/0004-6361/201935891
8	Post-Synthetic Modification of Metal–Organic Frameworks Towar...	Advanced Functional Mater...	2020	645	10.1002/adfm.202006291
9	Models and materials for generalized Kitaev magnetism	Journal of Physics Conden...	2017	613	10.1088/1361-648x/aa8cf5
10	Tropospheric Ozone Assessment Report: Present-day tropospheri...	Elementa Science of the A...	2018	592	10.1525/elementa.302

4 Citation network

The directed graph has 7,597 papers and 17,386 internal citation edges ("paper A cites paper B", both authored by our faculty). The raw `citation_network.csv` also carries 40,639 outgoing and 26,854 incoming edges to and from

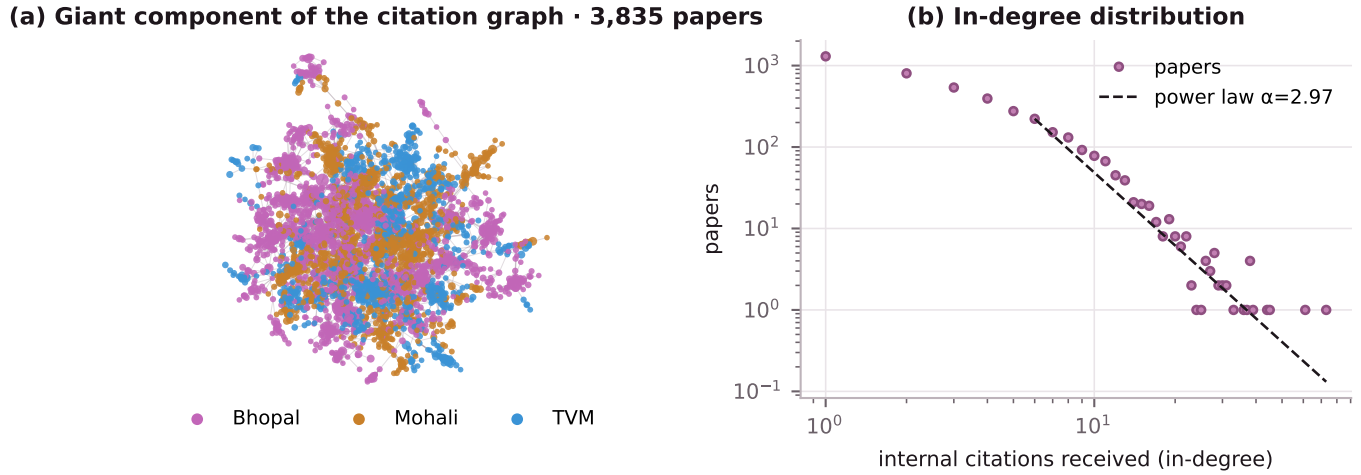


Figure 3: (a) Giant weakly connected component of the internal citation graph (nodes = papers, size ∝ PageRank, colour = institute). (b) In-degree distribution on log–log axes with the best power-law fit for reference.

other Indian institutes. About half the papers (51%) sit in one weakly connected component, and the median citation lag is 2 years. 96% of internal edges join two papers of the *same* institute: the three IISERs barely build on each other's work.

The in-degree distribution is heavy-tailed. A power law fits above $x_{\min} = 6$ with $\alpha = 2.97$, but a log-normal fits significantly better (likelihood ratio $R = -3.92$, $p < 0.001$), the usual signature of cumulative advantage tempered by ageing rather than pure preferential attachment. PageRank (Table 4) surfaces papers that are cited by *well-cited* internal papers: Bhopal's noncentrosymmetric-superconductor line and Mohali's atmospheric-chemistry papers outrank far more cited papers, because they seed long internal citation chains. The Planck 2018 entry ranks highly only because the papers in that series cite one another, which is why the dashboard offers a ranking without large collaborations.

Table 4: PageRank top 10 on the internal citation graph ($\alpha = 0.85$); *in* = citations from other corpus papers.

#	Title	Year	Inst.	In	All cites
1	Superconducting properties of the noncentrosymmetric supercondu…	2016	Bhopal	20	57
2	Time-reversal symmetry breaking in the noncentrosymmetric super…	2017	Bhopal	33	83
3	Overview of VOC emissions and chemistry from PTR-TOF-MS measure…	2016	Mohali	14	116
4	Contribution of post-harvest agricultural paddy residue fires i…	2016	Mohali	27	104
5	Strong interlayer coupling mediated giant two-photon absorption…	2016	Bhopal	28	71
6	Planck 2018 results	2020	TVM	73	14,257
7	Trace level detection of nitroanilines using a solution process…	2016	Bhopal	11	102
8	Crystal-to-Crystal Synthesis of Triazole-Linked Pseudo-proteins…	2016	TVM	44	80
9	Dynamical Phase Transition in Neutron Stars	2018	Bhopal	15	19
10	Effect of magnetic field on the burning of a neutron star	2018	Bhopal	6	7

5 Co-authorship network

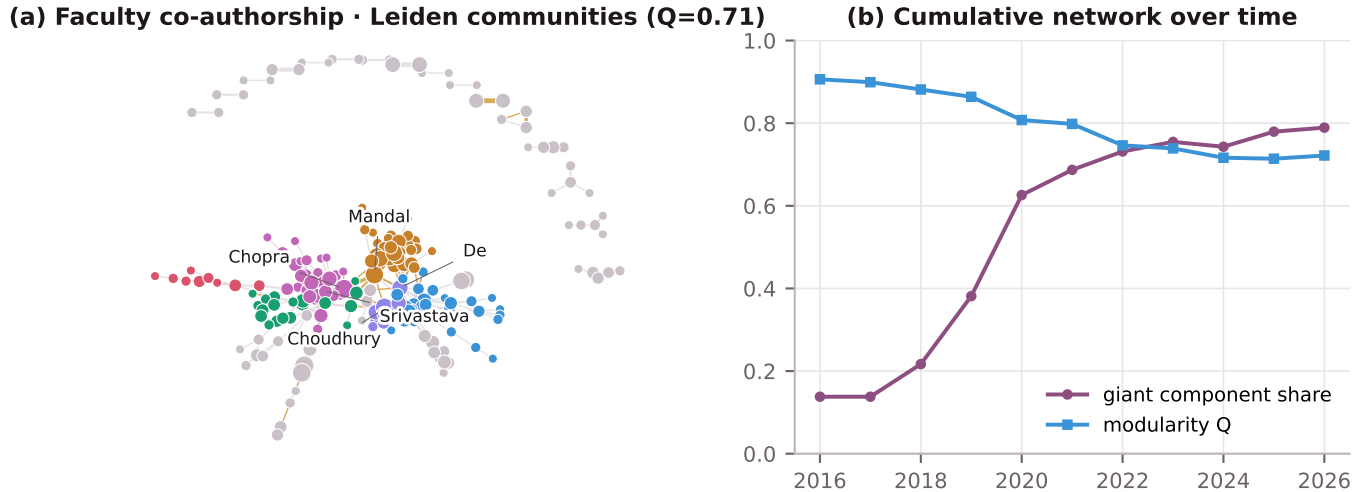


Figure 4: (a) Faculty co-authorship network (papers with ≤50 authors; node size = joint papers; gold edges cross institutes; the six largest Leiden communities coloured, the five strongest brokers named). (b) Cumulative network by year.

Nodes are our 397 matched faculty, joined when they co-authored a paper (weight = joint papers). Only 204 faculty have a co-author among the other faculty; the remaining 193 collaborate only with students, postdocs or outsiders. The 304 ties form a sparse (mean degree 3.0) but strongly clustered network. Its clustering of 0.19 is 7 times the configuration-model value (0.03) at a similar path length (4.99 vs 4.11), giving small-world $\sigma = 5.8$ (90% of 50 null graphs give 3.3–18.6, all well above 1). Leiden finds 27 communities with modularity $Q = 0.71$ (Louvain 0.79). They agree only partly with departments (NMI 0.63, ARI 0.32): the large communities are *interdisciplinary within one institute*, for example Bhopal chemistry with biology, and TVM chemistry with physics.

Brokers with the highest betweenness are Deepak Chopra (Chemistry, Bhopal), Sukhendu Mandal (Chemistry, TVM) and Aasheesh Srivastava (Chemistry, Bhopal) (Table 5). From 2016 to 2026 the giant component grew from 14% to 79% of connected faculty while modularity fell from 0.91 to 0.72 (Fig. 4b): the network is knitting together without losing its group structure. As a first model we predicted which faculty pairs would *start* collaborating after 2022 from the network up to 2022. Preferential attachment (AUC 0.71) and same-department membership (0.70) beat common-neighbour scores such as Adamic–Adar (0.64): new ties go to already well-connected people in the same department more than to friends of friends.

Table 5: Most collaborative (degree) and most bridging (weighted betweenness) faculty.

#	Degree	k	Betweenness	b
1	Deepak Chopra (Bhopal)	13	Deepak Chopra (Bhopal)	0.098
2	Santanu Kumar Pal (Mohali)	13	Sukhendu Mandal (TVM)	0.072
3	Sukhendu Mandal (TVM)	13	Aasheesh Srivastava (Bhopal)	0.042
4	Arijit Kumar De (Mohali)	10	Arijit Kumar De (Mohali)	0.039
5	Joy Mitra (TVM)	10	Angshuman Roy Choudhury (Mohali)	0.033
6	Ram Kumar Mishra (Bhopal)	9	Sanjay Singh (Mohali)	0.024
7	Aasheesh Srivastava (Bhopal)	9	Kausik Chattopadhyay (Mohali)	0.022
8	M M Shaijumon (TVM)	9	Sugumar Venkataramani (Mohali)	0.021

6 Indian connectivity

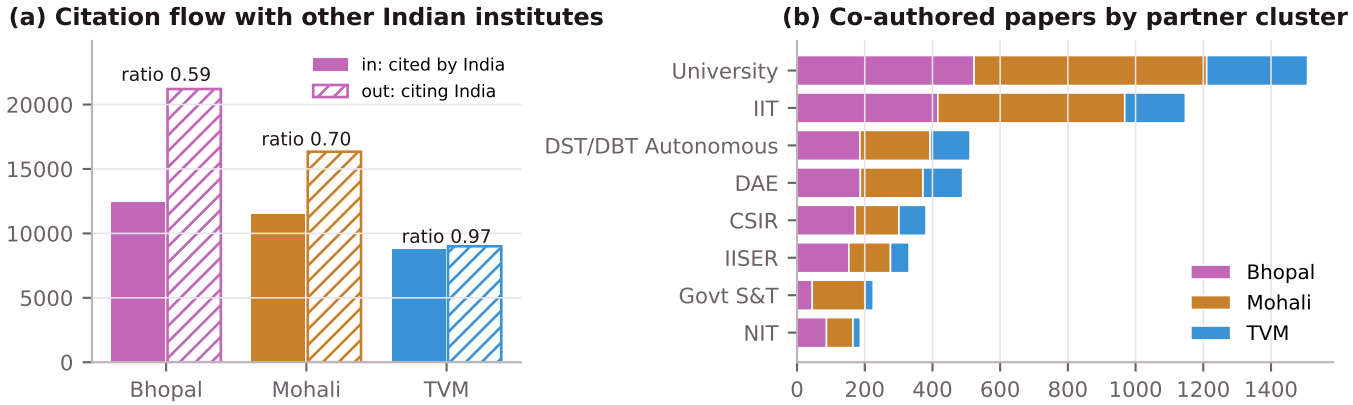


Figure 5: (a) Citations exchanged with other Indian institutes: in = an Indian-authored paper cites ours, out = our paper cites one; the in/out ratio is printed above each pair. (b) Co-authored papers by partner cluster.

Our faculty co-author with 542 other Indian institutes, and 2,033 Indian institutes are linked to them by co-authorship or citation. Bhopal received 12,444 citations from other Indian institutes and gave 21,207 (ratio 0.59); Mohali 11,485 vs 16,340 (0.70); TVM 8,745 vs 9,000 (0.97). The ratio is partly a time effect: references are complete at publication, while incoming citations keep arriving. The most frequent Indian citer of Bhopal and Mohali is the Academy of Scientific and Innovative Research (CSIR's PhD-granting academy), and of TVM the Homi Bhabha National Institute (DAE). The strongest co-authorship partners are IIT Kanpur (Bhopal), TIFR (Mohali) and IISc (TVM). By cluster, universities and IITs dominate shared papers, followed by DST/DBT autonomous institutes, DAE and CSIR; these *cross-cluster* links (IISER↔IIT, IISER↔CSIR/DAE) far outnumber IISER↔IISER papers. Topic overlap (Jaccard on the top-50 OpenAlex topics) with each partner is in `indian_connections.csv` and on the dashboard.

7 Why these findings matter

- **Raw counts mislead in exactly the places where rankings use them.** One Planck release decides which IISER looks "most cited". Any comparison of these institutes (NIRF-style or internal) should use field-weighted impact and report large-collaboration papers separately, as this dashboard does.
- **The IISER system is three parallel systems.** With 96% of mutual citations inside one institute and only 29 cross-IISER faculty ties, the three institutes work largely in isolation from each other. Their subfield portfolios overlap substantially (weighted Jaccard 0.52–0.61; dashboard: Research Areas), so the missing links are an

opportunity rather than a mismatch. Joint centres or shared PhD supervision could target the faculty pairs the link-prediction model ranks highest.

- **Collaboration is organised around people, not departments.** Communities cut across Chemistry, Biology and Physics inside each campus, and a few brokers (Table 5) hold large parts of the network together. That is a strength for interdisciplinary work and a fragility if those people leave.
- **The IISERs are absorbers of Indian science.** They cite universities, IITs and DAE/CSIR labs more than these cite them back. As the institutes mature (TVM is already near balance), the ratio is a simple indicator of whether they become sources of Indian knowledge rather than only consumers.

8 What the second round will add

- **Better community models.** A degree-corrected (nested) stochastic block model will test whether the institute/department structure is real or a modularity artefact, and core–periphery detection will find a collaborative core.
- **Temporal and disruption measures.** We will compute the CD disruption index and identify "sleeping beauty" papers on the citation graph, and fit a citation model with preferential attachment, ageing and fitness to explain the log-normal tail.
- **Representation learning for link prediction.** Node2vec embeddings plus topic similarity, evaluated on the same temporal split, to recommend cross-IISER collaborations where topics overlap but no ties exist (the three portfolios share a large part of their subfield mix).
- **A multilayer view.** Combine the citation, co-authorship and topic-similarity layers, and extract the disparity-filter backbone of the Indian partner network.